

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO2:** Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

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**QUESTIONS: 5**

**Total Marks:50**

#### Annexure A SENARIO BASED QUESTIONS

| <i>Criteria</i>  | Understanding of Scenario (2) | Identification of Key Issues (2) | Application of Concepts (2.5) | Decision Making (2) | Clarity of Response (1.5) | <i>Total(10)</i> |
|------------------|-------------------------------|----------------------------------|-------------------------------|---------------------|---------------------------|------------------|
| <i>Weightage</i> | 20%                           | 20%                              | 25%                           | 20%                 | 15%                       | <i>100%</i>      |
| <i>Q1</i>        |                               |                                  |                               |                     |                           |                  |
| <i>Q2</i>        |                               |                                  |                               |                     |                           |                  |
| <i>Q3</i>        |                               |                                  |                               |                     |                           |                  |
| <i>Q4</i>        |                               |                                  |                               |                     |                           |                  |
| <i>Q5</i>        |                               |                                  |                               |                     |                           |                  |

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#### ***Code of Conduct:***

*I G.Mohammad shareef/(192311288)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:G.MOHAMMAD SHAREEF***

### RUBRICS FOR CALCULATION:

| Criteria                     | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               |
|------------------------------|---------------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Understanding of Scenario    | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20%           | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |
| Application of Concepts      | 25%           | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |
| Decision Making              | 20%           | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |
| Clarity of Response          | 15%           | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |

**Annexure A**  
Scenario based Questions

## Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

### Sample Dataset

| Customer_ID | Age | Gender | Monthly_Spend (₹K) | Purchase_Frequency | Loyalty_Level |
|-------------|-----|--------|--------------------|--------------------|---------------|
| C1          | 23  | M      | 20                 | 5                  | Low           |
| C2          | NaN | Male   | 35                 | 8                  | Medium        |
| C3          | 31  | F      | NaN                | 10                 | High          |
| C4          | 105 | Female | 500                | 50                 | High          |
| C5          | 27  | male   | 25                 | NaN                | Low           |
| C6          | 40  | FEMALE | 45                 | 12                 | Medium        |
| C7          | 36  | M      | 38                 | 9                  | Low           |
| C8          | 42  | Female | 300                | 45                 | High          |

### Tasks

#### A. Handling Missing Values

- Identify missing values in **Age**, **Monthly\_Spend**, and **Purchase\_Frequency**
- Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

#### B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly\_Spend** using the **Z-score method**
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (**capping** / **transformation** / **removal**) and justify

#### C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male** / **Female**)

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## Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

### Sample Dataset

| Cust_ID | Age | Income (₹K) | Purchase_Value (₹K) | Credit_Score | Orders | Avg_Balance (₹K) | EMI (₹) | Customer_Class |
|---------|-----|-------------|---------------------|--------------|--------|------------------|---------|----------------|
| U1      | 24  | 40          | 100                 | 650          | 15     | 12               | 5000    | Low            |
| U2      | 34  | 60          | 150                 | 700          | 20     | 18               | 7000    | Medium         |
| U3      | 44  | 85          | 210                 | 720          | 28     | 25               | 9000    | Medium         |
| U4      | 52  | 120         | 320                 | 800          | 35     | 40               | 15000   | High           |
| U5      | 29  | 45          | 110                 | 680          | 18     | 14               | 5500    | Low            |
| U6      | 39  | 75          | 180                 | 710          | 24     | 22               | 8500    | Medium         |
| U7      | 56  | 150         | 360                 | 820          | 40     | 50               | 18000   | High           |
| U8      | 37  | 70          | 160                 | 705          | 22     | 20               | 8000    | Medium         |

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### Tasks

#### A. Covariance Analysis

- i. Compute covariance between **Income and Purchase\_Value** (show steps or partial calculation)
  - ii. Interpret whether the relationship is **positive or negative**
  - iii. Explain how covariance helps in identifying feature relationships
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#### B. Feature Selection

- i. Identify redundant attributes using **correlation or domain knowledge**
  - ii. Select an optimal subset of features for modeling
  - iii. Justify your selection
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#### C. Data Transformation

- i. Standardize the dataset (excluding **Cust\_ID** and **Customer\_Class**)
- ii. Explain why normalization/standardization is required

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## Q3. Data Transformation & Discretization in Education Analytics (*L3, L4, L5*)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

### Sample Dataset

| Student_ID | Study_Hours | Test_Score | Attendance (%) | Performance |
|------------|-------------|------------|----------------|-------------|
| S1         | 2           | 40         | 60             | Poor        |
| S2         | 3           | 50         | 65             | Average     |
| S3         | 4           | 55         | 70             | Average     |
| S4         | 5           | 65         | 75             | Good        |
| S5         | 6           | 75         | 80             | Good        |
| S6         | 7           | 85         | 85             | Good        |
| S7         | 8           | 90         | 88             | Excellent   |
| S8         | 9           | 95         | 90             | Excellent   |
| S9         | 10          | 98         | 92             | Excellent   |
| S10        | 11          | 100        | 95             | Excellent   |

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### Tasks

#### A. Equal-Width Binning

- Apply **3 bins** for Study\_Hours and Test\_Score
  - Calculate bin width and define intervals
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#### B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
  - Assign values accordingly
- 

#### C. Concept Hierarchy Construction

- Study\_Hours → Low / Medium / High

- Test\_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

## 5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

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### Sample Dataset: Visit Duration

**Customer\_ID Visit\_Duration (seconds)**

|     |      |
|-----|------|
| C1  | 15   |
| C2  | 30   |
| C3  | 45   |
| C4  | 60   |
| C5  | 120  |
| C6  | 300  |
| C7  | 600  |
| C8  | 900  |
| C9  | 1800 |
| C10 | 3600 |

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### Tasks

#### A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
  - Assign each value to its respective bin
  - Analyze how effectively this method groups short and long visit durations
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## B. Z-Score Normalization

- i. Compute the **mean and standard deviation** of visit duration
- ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
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## C. Min-Max Normalization

- i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- ii. Transform all values accordingly
- iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

## ANSWERS

### Q1. Data Preprocessing in Retail Analytics

#### A. Handling Missing Values

##### i. Missing Values

Customer Missing Attribute

C2        Age

C3        Monthly\_Spend

C5        Purchase\_Frequency

##### ii. Imputation Techniques

###### Method 1: Median Imputation

Age Values

23, 31, 105, 27, 40, 36, 42

Sorted:

23, 27, 31, 36, 40, 42, 105

Median = 36

So,

C2 Age = 36

###### Monthly Spend

20, 35, 500, 25, 45, 38, 300

Sorted

20, 25, 35, 38, 45, 300, 500

Median = 38

So,

C3 Monthly Spend = ₹38K

###### Purchase Frequency

5, 8, 10, 50, 12, 9, 45

Sorted

5, 8, 9, 10, 12, 45, 50

Median = 10

So,

C5 Purchase Frequency = 10

###### Method 2: k-NN Imputation

Nearest records are selected based on similar Age/Gender/Spend.

Example:

For C3, nearest customers are



- C6 (₹45K)
- C7 (₹38K)

Average  
 $= (45+38)/2$   
 $= ₹41.5K$

Similarly,  
 Age for C2  $\approx 35$  years  
 Purchase Frequency for C5  $\approx 9$

### iii. Justification

- Median is robust against outliers.
- k-NN preserves similarity among customers and gives better estimates for recommendation systems.

## B. Outlier Detection

### i. Z-score Method

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Age

Mean

$$= (23+36+31+105+27+40+36+42)/8$$

$$= 42.5$$

Standard deviation  $\approx 24.2$

For Age=105

$$Z = \frac{105 - 42.5}{24.2} = 2.58$$

Since  $|Z| > 2$ ,

Age=105 is an outlier.

### Monthly Spend

Mean

$$= (20+35+38+500+25+45+38+300)/8$$

$$= 125.1$$

Standard deviation  $\approx 169$

Spend=500

$$Z = \frac{500 - 125.1}{169} = 2.22$$

Outlier

Spend=300

$$Z \approx 1.03$$

Not an extreme outlier.

## ii. Extreme Records

Customer Outlier

C4      Age=105

C4      Spend=500

C8      Spend=300 (high but acceptable)

## iii. Treatment

Age 105

→ Cap at 80 years.

Spend 500

→ Winsorize (replace with upper percentile) or log transformation.

### Reason:

Removes abnormal influence while retaining records.

## C. Categorical Data Standardization

Original

M

Male

male

Female

F

FEMALE

### Standardized

Original    Standard

M            Male

Male        Male

male        Male

|          |          |
|----------|----------|
| Original | Standard |
| F        | Female   |
| Female   | Female   |
| FEMALE   | Female   |

## Q2. Covariance & Data Reduction

### A. Covariance Analysis

Income

40,60,85,120,45,75,150,70

Purchase Value

100,150,210,320,110,180,360,160

Mean Income=80.625

Mean Purchase=198.75

### Partial Calculation

For U1

[

$(40-80.625)(100-198.75)$

]

$=(-40.625)(-98.75)$

$=4011.7$

Similarly compute for all observations.

Total

$\approx 22,863$

Covariance

[

$=\frac{22863}{7}$

]

$\approx 3266$

### ii. Interpretation

Covariance is positive.

As income increases,  
purchase value also increases.

### iii. Importance

Covariance

- identifies relationships

- detects redundant variables
- supports dimensionality reduction

## **B. Feature Selection**

### **Highly correlated**

- Income
- Purchase\_Value
- Avg\_Balance
- EMI

### **Choose**

- Age
- Income
- Credit\_Score
- Orders

### **Target**

Customer\_Class

### **Reason**

These provide sufficient information with less redundancy.

## **C. Data Transformation**

Standardization Formula

$$Z = \frac{X - \mu}{\sigma}$$

Apply to

- Age
- Income
- Purchase\_Value
- Credit\_Score
- Orders
- Avg\_Balance
- EMI

### **Exclude**

- Cust\_ID
- Customer\_Class

### **Why Standardization?**

- Removes unit differences.
- Improves ML performance.
- Faster convergence.

- Important for KNN, PCA, SVM.

### **Q3. Data Transformation & Discretization**

#### **A. Equal Width Binning**

Study Hours

Minimum=2

Maximum=11

Range=9

Bin Width

$9/3=3$  Bins

| <b>Bin</b> | <b>Interval</b> |
|------------|-----------------|
| B1         | 2–5             |
| B2         | 5–8             |
| B3         | 8–11            |

#### **Assignments**

B1

2,3,4,5

B2

6,7,8

B3

9,10,11

#### **Test Score**

Minimum=40

Maximum=100

Range=60

Bin Width

20

Bins

40–60

60–80

80–100

Assignments

40,50,55

65,75

85,90,95,98,100

#### **B. Equal Frequency Binning**

10 records

≈3 records per bin

Study Hours

Bin1

2,3,4

Bin2

5,6,7

Bin3

8,9,10,11

**Test Scores**

Bin1

40,50,55

Bin2

65,75,85

Bin3

90,95,98,100

## **C. Concept Hierarchy**

**Study Hours**

**Hours Category**

2–4    Low

5–7    Medium

8–11   High

**Test Score**

**Score    Category**

40–59   Poor

60–79   Average

80–100   Excellent

#### **Q4. Normalization**

Data

200,300,400,600,1000

Minimum=200

Maximum=1000

##### **(a) Min-Max Normalization**

**Formula**

[

$$X' = \frac{X - 200}{800}$$

]

Value Normalized

200    0.000

300    0.125

400    0.250

600    0.500

1000   1.000

##### **(b) Z-score Normalization**

**Mean=500**

**Standard deviation≈282.84**

**Value Z-score**

200    -1.06

300    -0.71

400    -0.35

600    0.35

1000   1.77

#### **Q5. Online Shopping Behavior Analysis**

**Visit Duration**

**15,30,45,60,120,300,600,900,1800,3600**

##### **A. Equal Frequency Binning**

10 records

≈3 records/bin

Bin Values

B1 15,30,45

## Bin Values

B2 60,120,300

B3 600,900,1800,3600

## Analysis

- First bin contains short visits.
- Second bin contains medium visits.
- Third bin groups long visits, including extreme durations.

## B. Z-score Normalization

Mean

$$\mu = \frac{7470}{10} = 747$$

Standard deviation

$\approx 1096$

Formula

$$Z = \frac{X - 747}{1096}$$

Duration Z-score

15        -0.67

30        -0.65

45        -0.64

60        -0.63

120       -0.57

300       -0.41

600       -0.13

900       0.14

1800      0.96

3600      2.60

## Discussion

- Extreme values (1800 and 3600 seconds) have high Z-scores.
- Z-score highlights outliers without restricting the data to a fixed range.

## C. Min-Max Normalization

Formula



$$X' = \frac{X - 15}{3600 - 15}$$

Duration Min-Max Value

|      |       |
|------|-------|
| 15   | 0.000 |
| 30   | 0.004 |
| 45   | 0.008 |
| 60   | 0.013 |
| 120  | 0.029 |
| 300  | 0.080 |
| 600  | 0.163 |
| 900  | 0.247 |
| 1800 | 0.498 |
| 3600 | 1.000 |

### Comparison

#### Min-Max

Values are scaled to the range 0–1

Sensitive to minimum and maximum values

Useful for neural networks and distance-based algorithms

#### Z-score

Values are centered around 0 with unit standard deviation

Better for detecting outliers

Useful for statistical analysis and outlier detection